Simulation-Based Inference in Astrophysics

Bayesian inference when the simulator is easier than the likelihood

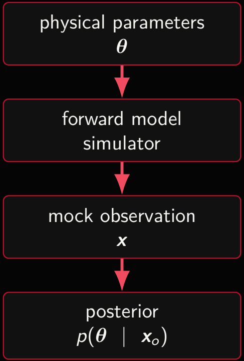
Python workshop **Forward models** **Likelihood-free inference**

Connor Bottrell

ICRAR - International Centre for Radio Astronomy Research

The Problem We Keep Meeting

- Astrophysics is full of inverse problems.
- We can often simulate the data-generating process.
- The corresponding likelihood may be unavailable, unstable, or too expensive.
- SBI asks: can simulations teach us the inverse map?



Workshop Goals

By the end, participants should be able to

- identify the prior, simulator, observation, and posterior target;
- explain what makes a likelihood hard to evaluate explicitly;
- train a neural posterior estimator from simulations;
- interpret posterior samples, degeneracies, and uncertainty;
- test the posterior using posterior predictive checks;
- explain why an accurate forward model is not optional.

Bayes' Theorem Is Still The Target

$$\boxed{p(\boldsymbol{\theta} \mid \boldsymbol{x}_o) = \frac{p(\boldsymbol{x}_o \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\boldsymbol{x}_o)}}$$

prior

What parameter values are plausible before seeing this data?

likelihood

How probable is the observed data under a candidate parameter value?

posterior

Which parameter values remain plausible after seeing the data?

What Makes The Likelihood Difficult?

A likelihood is not just a goodness-of-fit score

It is a normalized probability density for the observed data, including every stochastic step between parameters and measurement.

- Detector noise may be non-Gaussian, non-stationary, or poorly characterized.
- Selection effects can depend on detection pipelines and survey strategy.
- Nuisance variables may require high-dimensional marginalization.
- Observations may be indirect summaries rather than direct states.
- Simulations may include branching logic, thresholds, and numerical solvers.
- Some models are implicit: we can sample $\boldsymbol{x} \sim p(\boldsymbol{x} \mid \boldsymbol{\theta})$, but cannot evaluate $p(\boldsymbol{x} \mid \boldsymbol{\theta})$.

Astrophysical Examples

Likelihoods get awkward when

- the data are sparse Fourier samples;
- the noise model is learned empirically;
- a catalog was produced by a complex detection algorithm;
- the observable is a nonlinear summary statistic;
- the model includes subgrid physics or approximate numerics.

Common SBI candidates

- gravitational-wave strain recovery;
- VLBI black-hole imaging;
- galaxy population synthesis;
- strong lensing image analysis;
- cosmological fields and summaries;
- transient classification and rates.

Simulation-Based Inference

The key substitution

Instead of evaluating $p(\boldsymbol{x}_o \mid \boldsymbol{\theta})$, we generate many paired examples $(\boldsymbol{\theta}, \boldsymbol{x})$ and learn an inference object from them.

sample $\boldsymbol{\theta} \sim p(\boldsymbol{\theta})$ → simulate $\boldsymbol{x} \sim p(\boldsymbol{x} \mid \boldsymbol{\theta})$ → train neural density estimator → evaluate $p(\boldsymbol{\theta} \mid \boldsymbol{x}_o)$

The posterior is still Bayesian. The difference is how we obtain an approximation to it.

Three SBI Families

NPE

Neural posterior estimation learns

$$q_\phi(\boldsymbol{\theta} \mid \boldsymbol{x})$$

directly.

NLE

Neural likelihood estimation learns

$$q_\phi(\boldsymbol{x} \mid \boldsymbol{\theta})$$

then samples the posterior.

NRE

Neural ratio estimation learns a likelihood-to-evidence ratio

$$r_\phi(\boldsymbol{x}, \boldsymbol{\theta}).$$

In this workshop we use NPE: it is intuitive, amortized, and gives posterior samples quickly once trained.

Neural Posterior Estimation

Training objective

$$\hat{\phi} = \arg\max_{\phi} \sum_{i=1}^{N} \log q_{\phi}(\boldsymbol{\theta}_i \mid \boldsymbol{x}_i), \qquad \boldsymbol{\theta}_i \sim p(\boldsymbol{\theta}), \quad \boldsymbol{x}_i \sim p(\boldsymbol{x} \mid \boldsymbol{\theta}_i).$$

- The density estimator learns a conditional distribution, not a point estimate.
- It can represent degeneracies and non-Gaussian posterior shapes.
- After training, one model can be conditioned on many observations.
- The result is approximate and must be checked.

The Workflow In Python

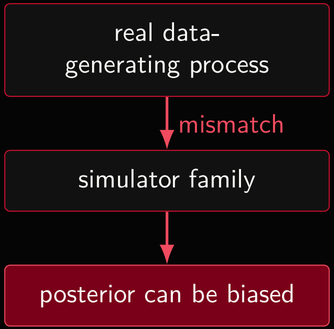
```
prior = BoxUniform(low, high)

theta = prior.sample((num_simulations,))
x = simulator(theta)

inference = NPE(prior=prior)
density_estimator = inference.append_simulations(theta, x).train()
posterior = inference.build_posterior(density_estimator)

samples = posterior.sample((5000,), x=x_observed)
```

Everything hangs on the simulator

The code is short because the scientific work is hidden inside the forward model: how parameters become mock observations.

The Forward Model Is The Scientific Claim

- SBI does not make the simulator true.
- It learns the posterior implied by the simulator and prior.
- If the simulator omits a relevant physical or instrumental effect, the posterior can be precise and wrong.
- Forward-model validation is therefore part of inference, not a separate housekeeping task.



What Must The Simulator Include?

Physical model

- source parameters;
- emission or waveform physics;
- cosmological and environmental effects;
- population-level variability.

Measurement model

- instrument response;
- noise and calibration uncertainty;
- selection function;
- data reduction and summaries.

good SBI = good prior + good simulator + good diagnostics

Priors Are Part Of The Model

- The training distribution is usually the prior.
- The neural estimator sees many more examples in high-prior-volume regions.
- A broad prior can waste simulations in irrelevant regions.
- A narrow prior can exclude the truth.
- Sequential SBI can concentrate simulations near a particular observation, but must handle proposal corrections carefully.

Workshop habit

Plot prior predictive simulations before training. If the simulations do not look like plausible observations, fix the simulator or the prior first.

Compression And Summary Statistics

The observation may be high-dimensional

Time series, images, spectra, catalogs, and maps can be much larger than the parameter vector.

- Use the full data when the density estimator can handle it.
- Use learned embeddings, summary networks, or physically motivated summaries when needed.
- Avoid summaries that erase information about parameters of interest.
- Test compression by checking posterior recovery on simulated data with known truth.

Diagnostics: Trust But Verify

Posterior diagnostics

- marginal and joint posterior plots;
- simulation-based calibration;
- coverage tests;
- sensitivity to simulation budget;
- prior sensitivity checks.

Posterior predictive checks

- draw $\boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta} \mid \boldsymbol{x}_o)$;
- simulate $\boldsymbol{x}^{(s)} \sim p(\boldsymbol{x} \mid \boldsymbol{\theta}^{(s)})$;
- compare simulated data to the observed data;
- look for systematic residual structure.

Worked Example: Noisy Chirp Recovery

Parameters

- chirp mass;
- amplitude;
- merger time;
- phase.

Observation

- one noisy strain segment;
- colored Gaussian noise;
- compact, inspectable simulator;
- posterior predictive waveform overlay.

This is deliberately not a production waveform. It is a transparent sandbox for learning SBI before moving to domain-grade tools.

Chirp Recovery As An SBI Graph

source parameters
$\boldsymbol{\theta}$

clean chirp
$h(t; \boldsymbol{\theta})$

detector-like noise
$n(t)$

strain segment
$x(t) = h(t) + n(t)$

posterior samples
$p(\boldsymbol{\theta} \mid x_o)$

Teaching moment

The posterior width is not just a neural-network artifact. It reflects noise, parameter degeneracy, prior support, and simulator adequacy.

Optional Extension: Sparse VLBI Ring

A different observation model

VLBI does not observe the image directly. It samples sparse Fourier components of the sky brightness distribution.

ring parameters → sky image → sparse Fourier samples → noisy visibilities

The SBI logic is unchanged, but the forward model now includes the measurement operator and the sparsity pattern.

Why Sparsity Matters

- Many images can agree with the same sparse visibility measurements.
- The posterior may constrain ring diameter while leaving asymmetry broad.
- Regularization, priors, and model class become scientifically visible.
- SBI can expose these degeneracies as posterior uncertainty rather than hiding them in one reconstruction.

Careful language

A toy ring model can teach sparse inverse problems. It should not be sold as reproducing the full Event Horizon Telescope imaging analysis.

Failure Modes

Scientific failure

- missing physics;
- wrong noise model;
- unmodeled selection;
- prior excludes plausible truth;
- summaries remove key information.

Computational failure

- too few simulations;
- poor neural density architecture;
- unstable normalization;
- extrapolation outside training support;
- no calibration check.

Practical Recipe

1. Write down $\boldsymbol{\theta}$, $\boldsymbol{x}$, and the posterior target.
2. Choose a prior with defensible support.
3. Build the simplest forward model that produces realistic mock data.
4. Plot prior predictive simulations.
5. Train an SBI estimator with a modest simulation budget.
6. Check inference on simulated observations with known truth.
7. Apply to the real observation.
8. Run posterior predictive checks and stress tests.

How To Read A Posterior Plot

- Narrow marginal: parameter is well constrained under the model.
- Wide marginal: data, prior, or simulator do not identify it strongly.
- Curved contour: nonlinear degeneracy.
- Boundary pile-up: prior edge or model mismatch may matter.
- Truth outside credible region in simulated tests: calibration problem.

Workshop phrase

The posterior is not a trophy. It is a compressed diagnostic of assumptions, data quality, and identifiability.

Live Notebook Structure

Notebook 1

- simulate noisy chirps;
- train NPE;
- infer one observation;
- overlay posterior predictive waveforms;
- add a glitch and diagnose mismatch.

Notebook 2

- simulate a toy ring image;
- measure sparse visibilities;
- infer ring parameters;
- visualize posterior image draws;
- vary uv sparsity.

Discussion Prompts

- What part of the forward model would you trust least?
- Which parameters are actually identifiable from the observation?
- How would you validate the simulator before using real data?
- What summaries would you choose if the raw data were too large?
- How would selection effects enter the simulator?
- What posterior predictive mismatch would make you stop?

Takeaways

The SBI mindset

- Start from Bayes, not from a neural network.
- Use simulations when likelihood evaluation is the hard part.
- Treat the simulator as the central scientific object.
- Prefer posterior distributions over point estimates.
- Check calibration and posterior predictive behavior.
- Be explicit about what the model can and cannot explain.

simulate forward **infer backward** **validate always**

Questions

What simulator would you trust enough to invert?

Bayes **SBI** **Astrophysics**